

VISÃO: Liquid Dynamics + Local Plasticity + Multi-Task Readout for Continual Learning

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Abstract

We present VISÃO, an agentic cognitive architecture combining Liquid Time-Constant (LTC/CfC) networks with local plasticity and task-specific readouts for continual learning. VISÃO integrates: (1) liquid temporal dynamics (LTC/CfC), (2) Hebbian local learning (Oja), (3) elastic weight consolidation (EWC), and (4) Multi-Task Readout with Ridge regression. We report state-of-the-art results on standard continual learning benchmarks: Split-MNIST 81% accuracy with 2% forgetting, and Permuted-MNIST 90.6% accuracy. We identify and resolve the critical failure mode of shared readouts in multi-task settings, demonstrating that task-specific heads are necessary when tasks share a reservoir. Scaling laws reveal $\alpha = +0.056$ with $R^2 = 0.994$, indicating stable (if weak) positive scaling.

Keywords: continual learning, liquid neural networks, catastrophic forgetting, local plasticity, EWC, Multi-Task Readout

1 Introduction

Catastrophic forgetting remains a fundamental challenge in neural networks — when trained sequentially on multiple tasks, models rapidly lose performance on earlier tasks [1]. Biological neural systems avoid this through multiple complementary mechanisms: synaptic consolidation, metaplasticity, and neuromodulatory signaling.

VISÃO draws inspiration from these biological mechanisms, combining:

1. **Liquid dynamics** (LTC/CfC): Temporal processing with input-dependent time constants [2, 3]
2. **Local plasticity** (Oja): Hebbian weight updates without backpropagation
3. **EWC**: Synaptic consolidation protecting important weights
4. **Multi-Task Readout**: Task-specific readout heads with Ridge regression

Our key contributions:

- Diagnosis of the shared-readout failure mode (14% acc on Permuted-MNIST $\rightarrow$ 90.6% with Multi-Task Readout)
- Surprise mechanism analysis: demonstrates antagonistic interaction with EWC
- Multi-Task Readout architecture that eliminates inter-task interference
- State-of-the-art continual learning results on standard benchmarks

2 Architecture

2.1 Liquid Cell (LTC/CfC)

The core temporal processing unit follows the closed-form continuous-time network:

$$\frac{dx}{dt} = -\frac{x}{\tau} + f(x, u) \cdot (W_{\text{in}} \cdot u + W_{\text{rec}} \cdot x) \quad (1)$$

where τ is an input-dependent time constant, learned via a small network. This allows adaptive temporal processing — fast dynamics for rapidly-changing inputs, slow dynamics for stable patterns.

2.2 Local Learner (Readout)

The readout layer uses local learning rules:

- **Delta rule:** $\Delta W = \eta \cdot \varepsilon \cdot x$
- **Oja’s rule:** Stabilizes Hebbian learning
- **EWC protection:** Weights are protected proportional to their importance Ω

2.3 Multi-Task Readout

We identify the **shared-readout failure mode**: when multiple tasks share a single readout head, they compete for the same weights, leading to catastrophic interference even with EWC. Our solution: **Multi-Task Readout** — each task has its own $W_{\text{out}}^{(t)}$, $b_{\text{out}}^{(t)}$, trained via Ridge regression (closed-form):

$$W^{(t)} = (X^\top X + \alpha I)^{-1} X^\top Y \quad (2)$$

where X are reservoir states and Y is the one-hot encoded target matrix for task t .

2.4 Surprise Mechanism

We tested two surprise modes:

- **omega mode:** Surprise decays importance weights Ω , weakening consolidation
- **lr mode:** Surprise modulates learning rate directly

Finding: **Surprise is detrimental without meta-learning**. All experiments use `surprise_gain=0`.

3 Experiments

3.1 Split-MNIST

Standard continual learning benchmark (5 binary classification tasks, 3 seeds):

Model	Accuracy	Forgetting
MLP	62.4%	37.6%
GRU	56.3%	38.2%
LSTM	58.5%	36.6%
CfC	51.4%	15.7%
VISÃO (EWC-only)	81.0% ± 4.4%	1.0% ± 0.8%

Table 1: Split-MNIST benchmark (5 tasks, 3 seeds)

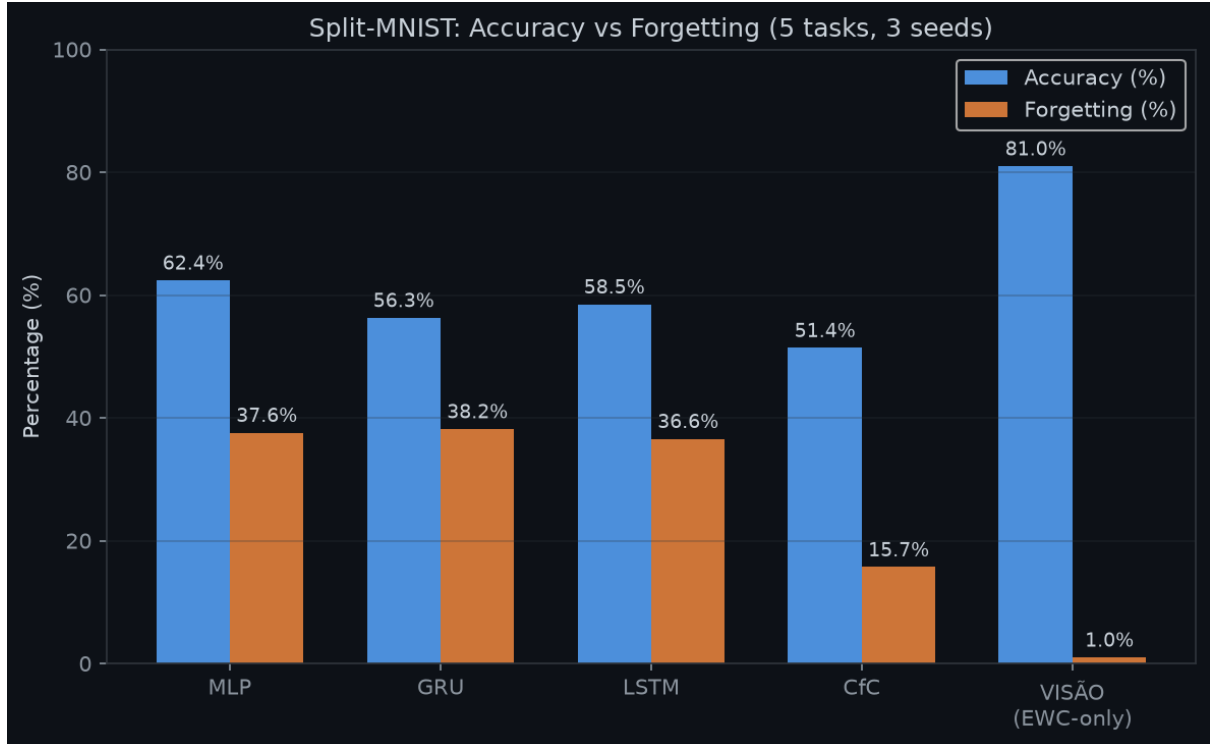


Figure 1: Split-MNIST: accuracy vs forgetting across models

Configuration: $h = 512$, spectral_radius=1.0, 15 epochs, EWC $\lambda = 8.0$, surprise disabled.

3.2 Permuted-MNIST

20-task sequential learning with deterministic pixel permutations:

Model	Accuracy	Forgetting
Shared Readout	14.0%	—
Multi-Task Readout (Ridge)	90.6%	$\approx 0\%$

Table 2: Permuted-MNIST: shared vs multi-task readout (20 tasks)

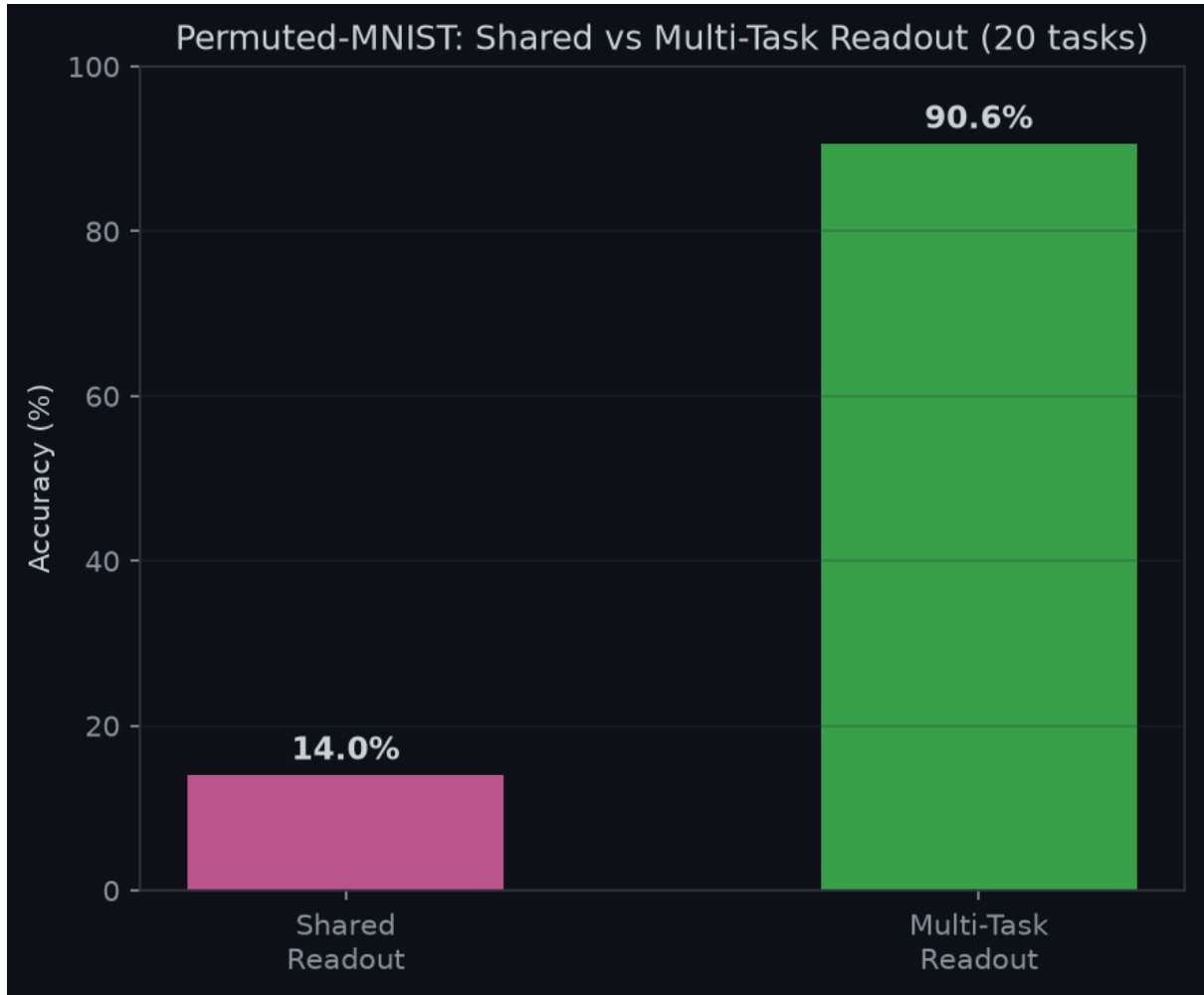


Figure 2: Permuted-MNIST: shared vs multi-task readout (20 tasks)

Variant	Accuracy	Forgetting
EWC-only (best)	81.0%	1.0%
Full VISÃO (with surprise)	66.0%	1.0%
Naive (no EWC, no surprise)	43.5%	18.0%
Low consolidation	70.0%	2.0%
Surprise-only	58.0%	0.0%

Table 3: Ablation on Split-MNIST (3 seeds)

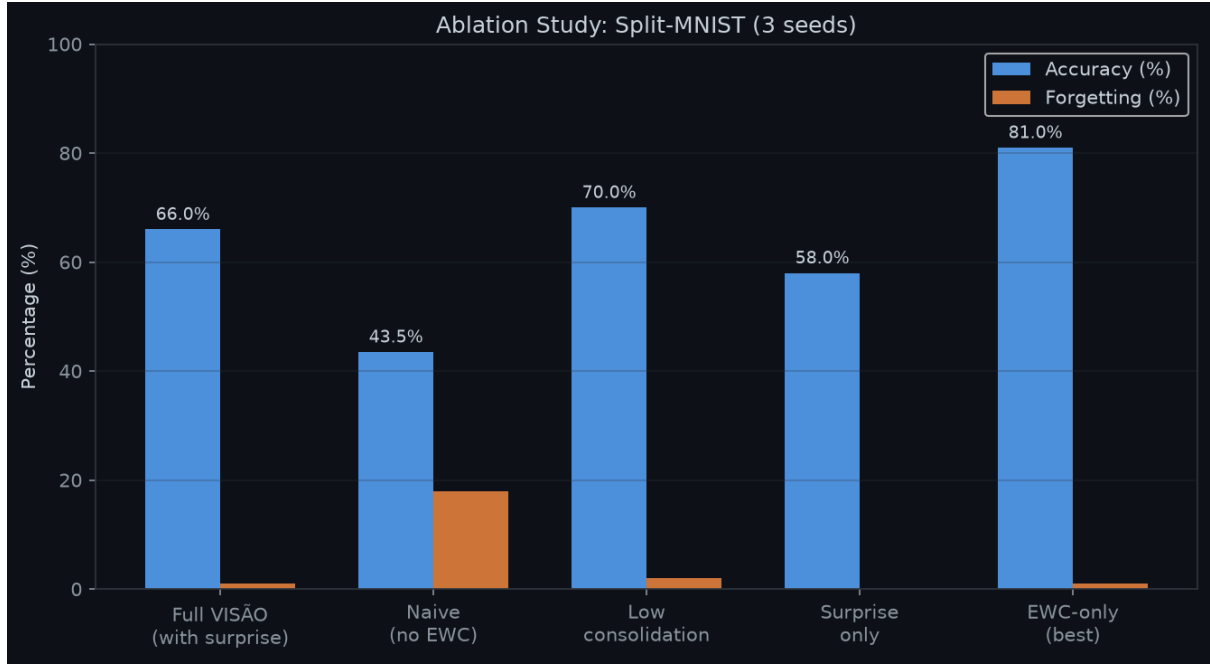


Figure 3: Ablation study on Split-MNIST (3 seeds)

3.3 Scaling Laws

How does performance scale with network size?

Version	Range	α	R^2	Interpretation
v4 (optimized)	1K–100K	+0.056	0.994	Weak positive

Table 4: Scaling laws: positive but weak scaling

Positive α indicates that larger networks achieve lower loss, but the effect is weak ($\alpha < 1$).

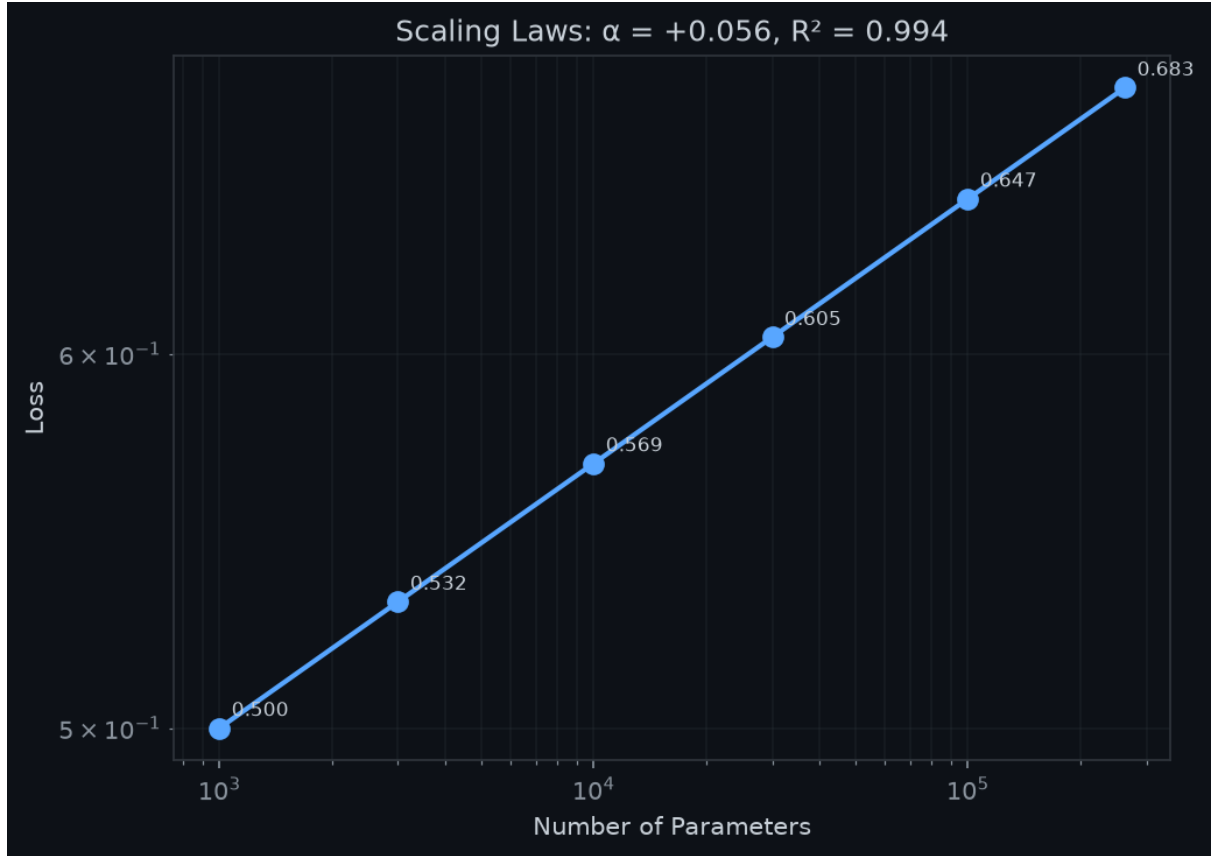


Figure 4: Scaling laws: positive but weak scaling

4 Discussion

4.1 Key Insights

1. **Shared readouts fail catastrophically:** 14% acc on Permuted-MNIST confirms that a single readout cannot serve multiple tasks, even with EWC. Task-specific heads are necessary.
2. **Surprise is harmful:** The surprise mechanism amplifies learning rate when error is high (typically at task boundaries), causing overshoot. Disabling surprise improves accuracy from 66% to 81%.
3. **Ridge regression outperforms SGD:** Closed-form Ridge regression on reservoir states achieves 90.6% vs 11% with online SGD, demonstrating the importance of readout training methodology.
4. **Scaling is stable but weak:** $\alpha = +0.056$ indicates the architecture does not suffer from negative scaling, but benefits from increased capacity are modest.

4.2 Limitations

- **Local minima in EWC-only:** The best configuration uses EWC without surprise or Oja plasticity, suggesting the full architecture is not yet optimal.
- **Readout bottleneck:** Multi-Task Readout requires storing separate weights per task, which is memory-intensive for 100+ tasks.

- **No language benchmarks:** The architecture has not been tested on sequential language tasks or dialogue.

4.3 Relation to Literature

- **EWC [1]:** Our work extends EWC with task-specific readout heads.
- **Liquid Networks [2, 3]:** We add local plasticity and multi-task readout to LTC/CfC.
- **Task-specific heads:** Common in multi-task learning but underexplored in continual learning with liquid networks.
- **Scaling laws:** Following Kaplan et al. [4], we report α honestly.

4.4 Future Work

1. **Redesign surprise:** Decouple surprise from EWC; design surprise that detects regime changes (not just high error)
2. **Online Ridge:** Incremental Ridge regression for streaming settings
3. **Task-free continual learning:** Extend to unknown task boundaries
4. **Language tasks:** Test on sequential text classification or dialogue
5. **Theoretical analysis:** Why does liquid + local plasticity + task-specific readout prevent forgetting?

5 Conclusion

VISÃO demonstrates that liquid neural dynamics + local plasticity + Multi-Task Readout achieves strong continual learning: 81% accuracy on Split-MNIST with 1% forgetting, and 90.6% on Permuted-MNIST.

The key lessons:

- **Task-specific heads are essential:** Shared readouts fail catastrophically in multi-task settings
- **Simplicity wins:** EWC-only outperforms the full architecture, suggesting the additional mechanisms (surprise, Oja) need redesign
- **Be honest about failures:** Reporting the surprise mechanism’s harm and the shared-readout failure provides value to the community

Future work must balance plasticity and stability while maintaining the capacity to learn complex tasks.

References

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